final-outline: Zeroth-Order Final Presentation Reading and Speaking Guide

This file is the final working outline for the 20-minute presentation on Nesterov and Spokoiny, **Random Gradient-Free Minimization of Convex Functions**. It is organized around the final overlay PDF, not the earlier blueprint or handout draft.

本文件是最终版汇报的工作大纲，基于当前的 **final__project__overlay.pdf**。它不是重新设计 slides，而是帮助三位组员明确：每页负责什么、原论文需要掌握到什么颗粒度、每页按 overlay 弹出顺序怎么讲。

---

0. Global presentation structure / 总体结构

The final PDF has **18 logical slides**. Slides 1-13 are the main 20-minute presentation. Slides 14-18 are backup slides for Q&A.

最终 PDF 有 **18 张逻辑页**。第 1-13 页为 20 分钟正式汇报，第 14-18 页为答疑备用页。

Speaker blocks / 汇报分块

Block	Speaker	Main slides	Time target	Core responsibility
Opening + hard theory	Leader	1-5	6.5-7.5 min	motivation, smoothing, oracle theory, acceleration
Nonsmooth + stochastic	Member A	6-8	4.0-4.5 min	constrained nonsmooth, finite-difference bias, stochastic extension
Smooth + experiment + conclusion	Member B	9-13	8.0-9.0 min	smooth baseline, nonconvex extension, Julia experiment, final takeaways
Backup Q&A	Assigned by topic	14-18	not in main timing	notation, identity proof, acceleration skeleton, function-call accounting, scaling summary

Overlay page mapping / overlay 页码映射

Logical slide	Final PDF overlay pages	Owner
1	1-2	Leader, all names visible
2	3-7	Leader
3	8-12	Leader
4	13-17	Leader
5	18-22	Leader
6	23-26	Member A
7	27-30	Member A

Logical slide	Final PDF overlay pages	Owner
8	31-35	Member A
9	36-39	Member B
10	40-43	Member B
11	44-48	Member B
12	49-53	Member B
13	54-58	Member B
14	59	Backup, shared notation
15	60	Backup, Leader
16	61	Backup, Leader
17	62	Backup, Member B
18	63	Backup, Member B

---

1. Reading depth by person / 每位组员的论文掌握颗粒度

1.1 Leader: highest difficulty block / 组长：最高难度理论部分

Core slides: 1-5, Backup B2-B3.
Main paper sections: Section 1.1, Section 2, Section 3, Section 6.
Paper page range: original PDF pp. 2-13 and pp. 22-26; journal pp. 528-540 and pp. 548-552.

Required mastery / 必须掌握 EN: The Leader should be able to explain why zeroth-order optimization is useful, why Gaussian smoothing is introduced, why the finite-difference oracle estimates the gradient of the smoothed objective, why the oracle variance has dimension dependence, and why acceleration retains the $1/k^2$ shape but pays a stronger dimension penalty.

中文：组长需要能解释四个核心问题：为什么需要 zeroth-order optimization；为什么引入 Gaussian smoothing；为什么有限差分随机 oracle 在期望意义下估计的是 f_μ 的梯度；为什么随机 oracle 的二阶矩带来维度因子；以及为什么 accelerated random search 保留 $1/k^2$ 但仍然有维度惩罚。

Specific paper targets / 具体阅读目标

Topic	Paper location	Formula numbers	Required depth
Motivation and problem setup	Sec. 1.1, original PDF pp. 2-5, journal pp. 528-531	(1), (2), (3), (5)	Understand motivation bullets and method form. Do not memorize literature review.
Gaussian smoothing definition	Sec. 2, original PDF pp. 6-8, journal pp. 532-534	(9), (10), (11)	Know what f_μ means and why convexity is preserved.
Bias bounds	Sec. 2, original PDF p. 8, journal p. 534	(18), (19), (20)	Know nonsmooth and smooth bias orders: $O(\mu L_0\sqrt{n})$, $O(\mu^2 L_1 n)$.
Gradient identity	Sec. 2, original PDF pp. 8-10, journal pp. 534-536	(21), (26)	Must be able to sketch the proof using change of variables and $E[u]=0$.
Oracles	Sec. 3, original PDF pp. 10-13, journal pp. 536-539	(30), (31)	Know g_μ, $\hat{g}_\mu$, g_0, and their roles.
Dimension factor	Sec. 3, original PDF pp. 10-13, journal pp. 536-539	(32), (34)-(37)	Must explain why $E\|g_0\|^2$ scales with $n+4$.

Topic	Paper location	Formula numbers	Required depth
Acceleration	Sec. 6, original PDF pp. 22-26, journal pp. 548-552	(60), (62), (63), (64)	Explain method skeleton and rate. Avoid detailed parameter algebra unless asked.

Reading method / 阅读方法 EN: Read Section 2 twice. First pass: definitions and theorem statements only. Second pass: derive the gradient identity in Backup B2 by hand. For Section 3, focus on the meaning of each oracle and Theorem 3. For Section 6, do not try to reproduce every parameter in Method FG_μ; instead, understand the structure: mix (x_k, v_k), query at y_k, update two sequences.

中文：Section 2 至少读两遍。第一遍只抓定义和定理结论，第二遍手推 Backup B2 中的 gradient identity。Section 3 重点读 oracle 的定义和 Theorem 3 的二阶矩含义。Section 6 不要硬背 FG_μ 的所有参数，而要掌握结构：由 (x_k, v_k) 混合出 y_k，在 y_k 查询随机 oracle，再同时更新 x_{k+1}, v_{k+1}。

Leader Q&A readiness / 组长答疑准备

1. Why Gaussian rather than coordinate directions?
   为什么用 Gaussian 随机方向，而不是坐标方向？
2. Is $g_\mu(x)$ an unbiased estimator of $\nabla f(x)$?
   $g_\mu(x)$ 是不是 $\nabla f(x)$ 的无偏估计？
3. Where does the dimension factor come from?
   维度因子从哪里来？
4. Why does acceleration become $O(n^2/k^2)$, not exactly $O(n/k^2)$?
   为什么 accelerated random search 是 $O(n^2/k^2)$ 形状，而不是简单的 $O(n/k^2)$？

1.2 Member A: nonsmooth and stochastic block / 组员 A：非光滑与随机优化部分

Core slides: 6-8.
Main paper section: Section 4.
Paper page range: original PDF pp. 14-19; journal pp. 540-545.

Required mastery / 必须掌握 EN: Member A should be able to explain how random search works under convex constraints, why projection is needed, how RS_0 differs from RS_μ, why finite differences introduce smoothing bias, and why stochastic zeroth-order optimization needs both a random direction u and a random sample ξ.

中文：组员 A 需要讲清楚 constrained nonsmooth convex optimization 中随机搜索如何工作，为什么需要投影，RS_0 和 RS_μ 的区别是什么，finite difference 为什么引入 smoothing bias，以及 stochastic zeroth-order optimization 中为什么同时有方向随机性 u 和样本随机性 ξ。

Specific paper targets / 具体阅读目标

Topic	Paper location	Formula numbers	Required depth
Nonsmooth constrained problem	Sec. 4, original PDF p. 14, journal p. 540	(38)	Know the problem statement and assumptions.
Projected random search	Sec. 4, original PDF pp. 14-15, journal pp. 540-541	(39)	Explain projection π_Q geometrically.
RS_0 rate	Sec. 4, original PDF pp. 15-16, journal pp. 541-542	(40), (42), (43)	State rate order and relation to subgradient method.

Topic	Paper location	Formula numbers	Required depth
RS_μ finite difference	Sec. 4, original PDF pp. 16-17, journal pp. 542-543	(45), (46)	Explain smoothing bias and $O(n^2/\epsilon^2)$-type finite-difference cost.
Stochastic objective	Sec. 4, original PDF pp. 17-18, journal pp. 543-544	(48), (49)	Know same-ξ finite differences.
Stochastic method	Sec. 4, original PDF pp. 18-19, journal pp. 544-545	(50), (51)	State expected rate shape and noise source.

Reading method / 阅读方法 EN: Read Section 4 by following the algorithms rather than the proofs. For each method, write down three items: problem, oracle, update. Then attach one rate statement. Proof details are less important than explaining the algorithmic logic and the cost difference between directional derivatives and finite differences.

中文：阅读 Section 4 时不要从证明入手，而是按“问题 - oracle - update”三件事组织。每种方法后面只配一个 rate statement。证明细节优先级低于算法逻辑和 RS_0、RS_μ 成本差异的解释。

Member A Q&A readiness / 组员 A 答疑准备

1. Why does nonsmooth optimization use projection?
   为什么非光滑约束问题需要投影？
2. Why is RS_0 theoretically cleaner than RS_μ?
   为什么 RS_0 理论上更干净？
3. What does smoothing bias mean?
   smoothing bias 是什么意思？
4. Why use the same stochastic sample ξ for both function evaluations?
   为什么两个函数值最好用同一个随机样本 ξ？

1.3 Member B: smooth baseline, nonconvex extension, experiment, closing / 组员 B：光滑基线、非凸扩展、实验与总结

Core slides: 9-13, Backup B4-B5.
Main paper sections: Section 5, Section 7, Section 8.1, Section 8.4.
Paper page range: original PDF pp. 19-22, pp. 26-29, pp. 36; journal pp. 545-548, pp. 552-555, p. 562.

Required mastery / 必须掌握 EN: Member B should be able to explain the smooth random gradient descent baseline, how it compares with ordinary gradient descent, what changes in the nonconvex setting, how the Julia experiment is constructed, how random blocks are counted, and why the experiment is qualitative support rather than a proof.

中文：组员 B 需要解释 smooth random gradient descent 的基线理论，它和普通 gradient descent 的差异，非凸情况下评价指标如何变化，Julia 实验如何构建，random blocks 如何计数，以及为什么实验只是支持“dimension-cost story”，不是证明。

Specific paper targets / 具体阅读目标

Topic	Paper location	Formula numbers	Required depth
Smooth problem	Sec. 5, original PDF p. 19, journal p. 545	(53)	Know smooth unconstrained setup.
RG_μ method	Sec. 5, original PDF p. 20, journal p. 546	(54), (55)	Know update and step size.

Topic	Paper location	Formula numbers	Required depth
Smooth rate	Sec. 5, original PDF pp. 20-22, journal pp. 546-548	(56), (57), (58), (59)	Explain $O(n/k)$ with smoothing error.
Nonconvex goal	Sec. 7, original PDF pp. 26-28, journal pp. 552-554	(65), (66), (67), (68), (69)	Explain stationary-point goal. No need for full proof.
Smooth quadratic chain	Sec. 8.1, original PDF pp. 28-29, journal pp. 554-555	(70), Table 1	Know function, optimizer availability, block accounting.
Experimental conclusion	Sec. 8.4, original PDF p. 36, journal p. 562	no formula	Explain: use random gradient-free only when gradients are unavailable or impractical.

Reading method / 阅读方法 EN: Read Section 5 with the goal of explaining the bridge from theory to the experiment. Read Section 7 only for the conceptual shift from objective gap to stationarity. Read Section 8.1 and 8.4 carefully because these pages support the experiment slide and the final conclusion. Keep a short code note: objective, algorithms, dimensions, seeds, metrics, and function-call accounting.

中文：阅读 Section 5 时重点看它如何连接 theory 和 experiment。Section 7 只需掌握从 objective gap 到 stationarity 的评价指标变化。Section 8.1 和 8.4 要细读，因为它们直接支撑实验页和最后结论。实验代码说明建议固定为六项：objective、algorithms、dimensions、seeds、metrics、function-call accounting。

Member B Q&A readiness / 组员 B 答疑准备

1. How is one random block counted?
   一个 random block 是怎么计数的？
2. Is a gradient evaluation comparable to one function call?
   一次 gradient evaluation 能不能等同于一次 function call？
3. Why does RG_0 track RG_μ in the experiment?
   为什么实验中 RG_0 和 RG_μ 很接近？
4. Why is the dimension-scaling table not a proof?
   为什么维度 scaling 表不是证明？
5. When should we not use random gradient-free methods?
   什么时候不应该使用 random gradient-free methods？

---

2. Slide-by-slide source map, mastery target, and overlay notes / 逐页来源、掌握目标与 overlay notes

The notes below are written in English for actual delivery, followed by Chinese intent notes. The speaker should not read everything mechanically; use these as controlled speaking prompts.

下面每页 notes 先给英文汇报讲法，再给中文意图说明。正式汇报时不需要逐字照读，但应按 overlay 弹出顺序表达。

Slide 1. Title and role split / 标题与分工

Owner: Leader opens; all members listed.
Final PDF overlay pages: 1-2.
Paper source: title page and abstract, original PDF p. 1, journal p. 527.
Key formulas: none.
Mastery target: Everyone should know the paper title, authors, and their own block.

Content task / 页面任务

EN: Set the scope: this is a 20-minute presentation about zeroth-order convex optimization using random Gaussian directions.

中文：开场明确汇报主题：只用函数值的 zeroth-order optimization，核心工具是 Gaussian random directions。

Overlay notes / 按 overlay 讲稿

Overlay 1 - title only

EN: "Our project is on *Random Gradient-Free Minimization of Convex Functions* by Yurii Nesterov and Vladimir Spokoiny. The central question is how far we can go in convex optimization when we do not use full gradients and only query function values."

中文：先点出论文和核心问题：不用完整梯度，只用函数值，优化还能做到什么程度。

Overlay 2 - role table appears

EN: "We organize the talk by responsibility. I will cover the main theoretical bridge: smoothing, random oracles, and acceleration. Member A will cover nonsmooth and stochastic settings. Member B will connect the smooth theory to our numerical experiment and close with takeaways."

中文：解释三人连续分块，避免交叉跳转。强调组长负责最难的理论桥梁。

Reading checklist / 阅读检查

- EN: Can each member summarize their block in one sentence?
- 中文：每个人是否能用一句话说明自己负责的部分？

Slide 2. Motivation: Why zeroth-order optimization? / 为什么需要 zeroth-order optimization?

Owner: Leader.
Final PDF overlay pages: 3-7.
Paper source: Sec. 1.1 Motivation, original PDF pp. 2-5, journal pp. 528-531. Read the motivation bullet list on journal pp. 528-529 and the random search scheme discussion on pp. 529-531.
Key formulas: problem (1), random finite-difference scheme (2), directional-derivative limit (3), smoothing definition preview (5).

Content task / 页面任务

EN: Introduce the black-box optimization problem and the cost question: what do we lose when gradients are unavailable?

中文：引入 black-box 情境，并提出本文主问题：不用梯度的代价是什么？

Overlay notes / 按 overlay 讲稿

Overlay 1 - optimization problem appears

EN: "We start from the standard problem $\min_{x\in\mathbb{R}^n} f(x)$. In a classical first-order course, the next object would usually be $\nabla f(x)$."

中文：从最熟悉的 unconstrained minimization 开始，建立和课程内容的连接。

Overlay 2 - classical methods need gradient

EN: "Classical gradient methods assume that the gradient is available, or at least that it can be computed reliably and cheaply."

中文：强调常规梯度法的隐含前提：梯度可得、可靠、成本可接受。

Overlay 3 - zeroth-order only queries function values

EN: "Zeroth-order methods remove that assumption. The algorithm only asks for numbers of the form $f(x)$, not the full vector $\nabla f(x)$."

中文：定义 zeroth-order：oracle 只返回函数值，不返回梯度向量。

Overlay 4 - application motivations + black-box diagram

EN: "This is useful when the objective is a black-box simulator, old legacy code, memory-heavy automatic differentiation, or a nonsmooth model where generalized gradients are hard to compute."

中文：对应论文 Section 1.1 的动机 bullets：旧软件、AD memory、程序员成本、非光滑梯度困难。

Overlay 5 - price question

EN: "The paper asks a precise complexity question: if we only use function values, how much slower are we? The answer is often: about a dimension factor, with more subtle behavior depending on smoothness."

中文：提出主线答案：通常多付一个维度因子，具体取决于 smooth/nonsmooth/stochastic/accelerated。

Mastery checklist / 掌握检查

- EN: Be able to state four motivations from Sec. 1.1 without reading the slide.
- EN: Be able to connect scheme (2) in the paper to the finite-difference oracle later.
- 中文：能不用看稿说出四个动机；能把论文公式 (2) 和后面 g_μ 联系起来。

Slide 3. Gaussian Smoothing: The Bridge to Gradients / Gaussian smoothing: 从函数值到梯度的桥

Owner: Leader.
Final PDF overlay pages: 8-12.
Paper source: Sec. 2, original PDF pp. 6-10, journal pp. 532-536.
Key formulas: smoothing definition (9), bias bounds (18), (19), gradient identity (21), symmetric identity (26).
Backup link: Backup B2 gives the derivation skeleton.

Content task / 页面任务

EN: Explain that random finite differences are not directly estimating ∇f in every setting. They estimate ∇f_μ, the gradient of a Gaussian-smoothed objective.

中文：解释关键桥梁：有限差分随机向量在期望意义下估计的是 f_μ 的梯度，而不一定是原函数 f 的梯度。

Overlay notes / 按 overlay 讲稿

Overlay 1 - smoothing definition

EN: "The paper defines the Gaussian-smoothed objective as $f_\mu(x) = E_u[f(x + \mu u)]$, where $u \sim N(0, I)$. This averages function values around x."

中文：先定义 f_μ：在 x 附近加 Gaussian 扰动后取平均。

Overlay 2 - gradient identity

EN: "The key identity is that $\nabla f_\mu(x)$ equals the expectation of a finite-difference random vector. This is the bridge from function values to a gradient-like object."

中文：强调这是全文最关键公式：有限差分 oracle 的期望就是 smoothed objective 的梯度。

Overlay 3 - finite differences estimate ∇f_μ

EN: "So the method is not magic: it is doing gradient descent on a smoothed surrogate, in expectation. The target is ∇f_μ, not always ∇f."

中文：澄清：不是凭空得到梯度，而是在平滑后的函数上做随机梯度估计。

Overlay 4 - nonsmooth and smooth bias bounds

EN: "The price of smoothing is bias. If f is Lipschitz but nonsmooth, the gap is bounded by $\mu L_0\sqrt{n}$. If f is smooth, the gap improves to order $\mu^2 L_1 n$."

中文：讲 smoothing bias。非光滑是 $O(\mu\sqrt{n})$，光滑是 $O(\mu^2 n)$。这也是为什么 smooth 情况更容易。

Overlay 5 - plot of $|x|$ smoothing

EN: "The plot gives the intuition. For $f(x) = |x|$, smoothing rounds the kink. Larger μ gives a smoother function, but it also moves farther from the original objective."

中文：用图解释：$|x|$ 的尖点被抹平；μ 越大越平滑，但偏差也越大。

Mastery checklist / 掌握检查

- EN: Can derive the identity in Backup B2 at a high level: change variables, differentiate under the integral, subtract $f(x)$ because $E[u] = 0$.
- EN: Can explain the difference between smoothing and finite differencing.
- 中文：能解释 μ 的 bias-stability tradeoff；能回答 "这个 oracle 估计的是谁的梯度"。

Slide 4. Random Oracles and the Dimension Factor / 随机 oracle 与维度因子

Owner: Leader.
Final PDF overlay pages: 13-17.
Paper source: Sec. 3, original PDF pp. 10-14, journal pp. 536-540.
Key formulas: oracle definitions (30), expectation/subgradient relation (31), second-moment bound (32), finite-difference moment bounds (34)-(37).
Backup link: Backup B1 notation.

Content task / 页面任务

EN: Define the three oracles and explain why randomness is useful only in expectation. Use the second-moment bound to introduce the dimension cost.

中文：定义三种 oracle，并用二阶矩 bound 引出维度因子。

Overlay notes / 按 overlay 讲稿

Overlay 1 - Euclidean notation and g_μ

EN: "The paper states everything in a general geometry with a matrix B. For the presentation, we use the Euclidean case $B = I$. The first oracle is the forward finite-difference oracle g_μ."

中文：说明 slides 简化为 $B = I$，避免听众被 general geometry 分散注意力。

Overlay 2 - symmetric oracle $\hat{g}_\mu$

EN: "The second oracle is the symmetric finite-difference version. It uses two shifted values and often has smaller bias in smooth settings, but costs another function evaluation."

中文：说明 symmetric finite difference 的优点和代价：偏差较小，但要两个 shifted function values。

Overlay 3 - directional derivative oracle g_0

EN: "The third oracle uses a directional derivative. It is not purely function-value-only, but it is theoretically clean and useful as a limiting or diagnostic object."

中文：解释 g_0 不完全是 zero-order function value oracle，但理论上更干净，也可作为 finite difference 的极限/诊断版本。

Overlay 4 - second moment bound

EN: "The important bound is $E\|g_0(x)\|^2 \le (n+4)\|\nabla f(x)\|^2$. This shows that the random direction estimate has a second moment that grows with dimension."

中文：讲核心式子：二阶矩多了 $n+4$，这就是后面复杂度多维度因子的来源。

Overlay 5 - random direction diagram and bullets

EN: "The direction is not guaranteed to be useful at each single step. It becomes useful after averaging over random directions. The dimension-dependent second moment is what creates the typical dimension cost."

中文：强调单次随机方向可能不好，但期望方向有意义；方差/二阶矩随维度增长。

Mastery checklist / 掌握检查

- EN: Can distinguish g_μ, $\hat{g}_\mu$, and g_0 without notes.
- EN: Can explain why $n+4$ appears conceptually, without deriving all Gaussian moments.
- 中文：能回答"为什么随机方法通常比梯度法慢 $O(n)$"。

Slide 5. Accelerated Random Search / 加速随机搜索

Owner: Leader.
Final PDF overlay pages: 18-22.
Paper source: Sec. 6, original PDF pp. 22-26, journal pp. 548-552.
Key formulas: Method FG$_\mu$ (60), parameter relations (61), Theorem 9 (62), complexity estimates (63), finite-difference parameter bound (64).
Backup link: Backup B3.

Content task / 页面任务

EN: Present acceleration at the structure and rate level, not at the parameter-algebra level. The main message is: acceleration is possible, but it does not remove the dimension penalty.

中文：只讲结构和 rate，不讲复杂参数。主旨：可以加速，但维度惩罚仍在。

Overlay notes / 按 overlay 讲稿

Overlay 1 - algorithm skeleton diagram

EN: "The accelerated method keeps two sequences, a current point x_k and an auxiliary point v_k. It mixes them into y_k, queries the random oracle, and updates both sequences."

中文：用 diagram 讲算法骨架：$(x_k, v_k) \to y_k \to g_\mu(y_k) \to (x_{k+1}, v_{k+1})$。

Overlay 2 - accelerated rate

EN: "The expected objective gap has the accelerated shape $1/k^2$, but with a zeroth-order dimension factor: roughly $O(n^2 L_1 R^2/k^2)$, plus smoothing error."

中文：强调 rate 形状：$1/k^2$ 保留，但系数有 n^2。

Overlay 3 - three bullets

EN: "This is the zeroth-order analogue of Nesterov acceleration. The shape is still accelerated, but random finite differences make the dimension dependence worse."

中文：对听众总结：这是 Nesterov acceleration 的 random oracle 版本。

Overlay 4 - dimension penalty bullet emphasis

EN: "The important caveat is that acceleration improves dependence on k, not the fact that we pay for using random directions."

中文：明确 acceleration 改善迭代次数的 k 依赖，但不是消除维度代价。

Overlay 5 - comparison table

EN: "This table is the clean comparison: full gradient methods have $O(1/k)$ and $O(1/k^2)$; random gradient-free versions keep the same shapes but become $O(n/k)$ and $O(n^2/k^2)$."

中文：用表格收束组长部分，同时为 Member A/B 的其他 rate 做铺垫。

Mastery checklist / 掌握检查

- EN: Can state Method FG_μ structure from memory.
- EN: Can explain why the main slide hides parameter algebra.
- 中文：如果教授追问细节，应把回答引到 Backup B3，而不是现场推所有参数。

Transition to Member A / 转场给组员 A

EN: "So far we saw the smoothing and oracle machinery. Member A will now show how the same machinery is used for nonsmooth convex and stochastic problems."

中文：转场时强调：前面是工具，后面是不同问题类别的应用。

Slide 6. Nonsmooth Convex Optimization / 非光滑凸优化

Owner: Member A.
Final PDF overlay pages: 23-26.
Paper source: Sec. 4, original PDF pp. 14-16, journal pp. 540-542.
Key formulas: problem (38), Method RS_μ (39), RS_0 analysis (40), rate (42), complexity (43).

Content task / 页面任务

EN: Explain projected random search for nonsmooth convex constrained problems.

中文：讲清楚非光滑有约束问题中的 projected random search。

Overlay notes / 按 overlay 讲稿

Overlay 1 - constrained nonsmooth problem

EN: "Now the problem is $\min_{x\in Q} f(x)$, where f is convex but may be nonsmooth. The feasible set Q is important, because a random step can leave the feasible region."

中文：说明问题从 unconstrained smooth 转到 constrained nonsmooth，随机步可能跑出可行域。

Overlay 2 - projected update

EN: "The update is a projected step: take a random gradient-free step and then project back to Q. This is analogous to projected subgradient descent."

中文：解释公式：先沿 g_μ 随机方向走，再用 π_Q 投回可行集。

Overlay 3 - projection diagram

EN: "The picture shows this geometrically. The raw point $x_k - hg_\mu(x_k)$ may lie outside Q, so the next iterate is its projection."

中文：用图讲 projection，不需要推投影性质。

Overlay 4 - RS_0 rate and interpretation

EN: "For the directional-derivative version RS_0, the expected gap has the same spirit as subgradient descent, but with a $\sqrt{n}$-type dimension dependence in the rate statement."

中文：讲 rate：和 projected subgradient descent 类似，但随机方向引入维度相关项。

Mastery checklist / 掌握检查

- EN: Can define Q, π_Q, and $\hat{x}_N$.
- EN: Can state why projection is needed.
- 中文：能解释 RS_0 是 directional-derivative 版本，不是实际最严格的 function-value-only 版本。

Slide 7. Finite Differences and Smoothing Bias / 有限差分与 smoothing bias

Owner: Member A.
Final PDF overlay pages: 27-30.
Paper source: Sec. 4, original PDF pp. 16-17, journal pp. 542-543; also Sec. 2 bias bound on original PDF p. 8, journal p. 534.
Key formulas: bias bound (18), RS_μ theorem (45), parameter choice and complexity (46).

Content task / 页面任务

EN: Compare RS_0, RS_μ, and subgradient methods. Explain why true zero-order finite differences pay more in nonsmooth settings.

中文：比较 RS_0、RS_μ 和 subgradient method；解释 true zero-order finite difference 在非光滑情况下更贵。

Overlay notes / 按 overlay 讲稿

Overlay 1 - method comparison table

EN: "This table separates three objects. RS_0 uses a directional derivative and gives clean theory. RS_μ uses forward finite differences and is the true function-value method. The subgradient method is first-order in the nonsmooth sense because it needs a subgradient."

中文：用表格建立三者区别：RS_0 理论清楚，RS_μ 真正只用函数值，subgradient 需要次梯度。

Overlay 2 - smoothing bias formula

EN: "Finite differences with $\mu > 0$ are tied to the smoothed objective. The bias at the optimum can be bounded by $\mu L_0 \sqrt{n}$."

中文：解释 formula：有限差分用 f_μ，所以最优值附近有 smoothing bias。

Overlay 3 - complexity comparison

EN: "The paper summarizes the nonsmooth cost as RS_0 having an $O(n/\epsilon^2)$-type complexity, while the finite-difference RS_μ version can increase to $O(n^2/\epsilon^2)$."

中文：强调非光滑 finite-difference 版本比 directional derivative 版本更贵。

Overlay 4 - tuning bullets

EN: "A smaller μ reduces smoothing bias, but if it is too small, finite differences can suffer numerical cancellation. The theory gives safe bounds; empirical tuning can behave differently."

中文：讲 μ 的实际调参逻辑：太大 bias 大，太小数值消去风险高。

Mastery checklist / 掌握检查

- EN: Can say why RS_μ is more relevant for black-box function-value code.
- EN: Can explain the bias-versus-cancellation tradeoff in plain language.
- 中文：能说明为什么 finite difference setting 的复杂度可能变成 $O(n^2/\epsilon^2)$。

Slide 8. Stochastic Zeroth-Order Optimization / 随机 zeroth-order 优化

Owner: Member A.
Final PDF overlay pages: 31-35.
Paper source: Sec. 4 stochastic part, original PDF pp. 17-19, journal pp. 543-545.
Key formulas: stochastic objective (48), stochastic oracles (49), Method SS_μ (50), theorem bound (51). The abstract also states the expected rate shape $O(n/k^{1/2})$.

Content task / 页面任务

EN: Extend the zero-order idea from deterministic $f(x)$ to stochastic objectives $E_\xi[F(x,\xi)]$.

中文：把确定性函数值 oracle 扩展到随机目标函数 $E_\xi[F(x,\xi)]$。

Overlay notes / 按 overlay 讲稿

Overlay 1 - stochastic objective

EN: "In stochastic optimization, the objective itself is an expectation: $f(x) = E_\xi[F(x,\xi)]$. This matches simulation or sample-based problems."

中文：先解释 stochastic objective 的含义：目标函数本身是样本平均/期望。

Overlay 2 - stochastic finite-difference oracle

EN: "The zero-order stochastic oracle uses both a random direction u and a random sample ξ. The finite difference compares two function values under the same sample when possible."

中文：说明 oracle 中有两类随机性：方向 u 和样本 ξ。

Overlay 3 - workflow starts with sample ξ

EN: "The workflow first draws ξ. This fixes the stochastic environment for the finite difference."

中文：强调先抽样本环境，后在同一环境下比较两个函数值。

Overlay 4 - full workflow

EN: "Then we sample a direction, make same-ξ queries, and update x. Reusing ξ reduces sampling noise in the difference."

中文：讲完整流程：sample ξ，sample u，same-ξ queries，update。

Overlay 5 - rate and randomness bullets

EN: "The expected rate shape is stochastic-approximation-like, with a dimension factor. The important modeling point is that noise now comes from both the random direction and the stochastic sample."

中文：讲结论：rate 类似 stochastic approximation，但带维度因子；噪声来自 u 和 ξ。

Mastery checklist / 掌握检查

- EN: Can define $F(x,\xi)$ versus $f(x)$.
- EN: Can explain why same-ξ comparison reduces variance.
- 中文：能回答"如果无法用同一个 ξ 怎么办"：论文也讨论 directional stochastic oracle 或其他实现误差模型，但主 slide 不展开。

Transition to Member B / 转场给组员 B

EN: "The next part returns to smooth problems. Member B will show the basic smooth random gradient descent result and then connect it to our numerical experiment."

中文：转场说明：从非光滑/随机扩展回到 smooth convex baseline 和实验。

Slide 9. Smooth Convex Baseline: Random Gradient Descent / 光滑凸基线：随机梯度下降

Owner: Member B.
Final PDF overlay pages: 36-39.
Paper source: Sec. 5, original PDF pp. 19-22, journal pp. 545-548.
Key formulas: smooth problem (53), Method RG_μ (54), step size (55), Theorem 8 bound (56), finite-difference parameter (58).

Content task / 页面任务

EN: Present the smooth convex baseline and make the direct comparison with ordinary gradient descent.

中文：给出 smooth convex baseline，并与普通 gradient descent 对比。

Overlay notes / 按 overlay 讲稿

Overlay 1 - smooth problem

EN: "Now assume f is smooth, specifically with Lipschitz-continuous gradient. This is the setting closest to standard gradient descent."

中文：说明这里是 $C^{1,1}$ 光滑函数，最接近课程里的 gradient descent。

Overlay 2 - RG_μ update and step size

EN: "The random method has the same structure as gradient descent, except $\nabla f(x_k)$ is replaced by the random finite-difference oracle $g_\mu(x_k)$. The step size is scaled by $1/(nL_1)$."

中文：更新式和 GD 类似，但梯度换成 g_μ，步长中出现 $n+4$。

Overlay 3 - rate statement

EN: "The expected error has an $O(nL_1R^2/N)$ term plus a smoothing-error term. Compared with ordinary gradient descent, the $1/N$ shape remains, but the dimension factor appears."

中文：讲 rate：仍然是 $1/N$ 型，但多了 n，另外有 μ 带来的 smoothing error。

Overlay 4 - bullets and experiment bridge

EN: "This slide is the theory bridge to our experiment. The experiment tests exactly this story: random finite differences can descend, but full gradients should be much faster when they are available."

中文：把理论和实验连接起来：RG 可以下降，但如果梯度可用，GM 应该更快。

Mastery checklist / 掌握检查

- EN: Can state the difference between $O(1/k)$ and $O(n/k)$ in words.
- EN: Can explain the smoothing-error term qualitatively.
- 中文：能说明为什么这个 slide 是后面 quadratic-chain experiment 的理论依据。

Slide 10. Nonconvex Extension: What Changes? / 非凸扩展：变化在哪里？

Owner: Member B.
Final PDF overlay pages: 40-43.
Paper source: Sec. 7, original PDF pp. 26-28, journal pp. 552-554.
Key formulas: nonconvex problem (65), Method $\widehat{RS}_\mu$ (66), descent inequality (67), expected gradient-norm bound (68), nonsmooth smoothed-stationarity bound (69).

Content task / 页面任务

EN: Explain that in nonconvex problems we no longer promise global optimality; we aim for stationarity.

中文：说明非凸情况下不能保证全局最优，评价指标转为 stationarity。

Overlay notes / 按 overlay 讲稿

Overlay 1 - nonconvex problem

EN: "The paper also discusses nonconvex objectives. Here the problem statement looks similar, but the guarantee must change."

中文：引入非凸扩展，强调问题形式相似但保证不同。

Overlay 2 - stationarity goal

EN: "For nonconvex functions, a small objective gap to the global optimum is not the right target. Instead, the goal is to make $\|\nabla f(x)\|$ small, meaning a stationary point."

中文：解释目标从 objective gap 改为 gradient norm/stationary point。

Overlay 3 - saddle and local minimum illustration

EN: "The picture shows why. A point with zero gradient may be a saddle or a local minimum, and a local minimum need not be global."

中文：用图帮助非优化背景听众理解非凸困难。

Overlay 4 - bullets

EN: "So the analysis studies expected gradient norm, or the gradient of the smoothed objective f_μ. Smoothing is again useful because it gives a differentiable surrogate."

中文：总结：分析对象是 ∇f 或 ∇f_μ 的范数，而不是 $f(x) - f^*$。

Mastery checklist / 掌握检查

- EN: Can explain stationary point without overclaiming global optimality.
- EN: Can say why smoothing is useful in nonsmooth nonconvex settings.
- 中文：注意不要说 "nonconvex 方法找到 global minimum"。只能说 stationarity。

Slide 11. Smooth Quadratic Experiment: Setup / 光滑二次实验：设置

Owner: Member B.
Final PDF overlay pages: 44-48.
Paper source: Sec. 8.1, original PDF pp. 28-29, journal pp. 554-555.
Key formulas: smooth quadratic chain (70). Original paper uses $n = 256$ and notes $L_1(f_n) \le 4$, $R^2 \le (n+1)/3$. Our presentation uses $n \in \{32, 64, 128\}$ and seeds 1:10.

Content task / 页面任务

EN: Define the Julia experiment as a reproduction-style small experiment based on the paper' s smooth quadratic chain.

中文：说明实验是基于论文 Section 8.1 的 quadratic chain 做的小规模 Julia reproduction。

Overlay notes / 按 overlay 讲稿

Overlay 1 - quadratic chain function

EN: "For the experiment, we use the smooth quadratic chain from Section 8.1. It is a standard test function for smooth first-order methods."

中文：先讲目标函数来源：论文 Sec. 8.1 的 quadratic chain。

Overlay 2 - known optimizer and computable gaps

EN: "Because this is a quadratic problem with a known optimizer, we can compute objective gaps directly. That makes the plots and success thresholds meaningful."

中文：强调 known optimizer 的好处：objective gap 可精确计算。

Overlay 3 - methods

EN: "We compare full-gradient GM, finite-difference random gradient RG_μ, and diagnostic RG_0. RG_0 is not the true function-value method; it checks whether finite-difference bias is small."

中文：明确三种方法，尤其说明 RG_0 是诊断基准，不是真正 function-value-only。

Overlay 4 - dimensions and seeds

EN: "We run dimensions 32, 64, and 128, with seeds 1 through 10. This gives a small but reproducible dimension-scaling check."

中文：讲实验规模：三组维度、10 个随机种子。

Overlay 5 - metrics and μ

EN: "Metrics are relative objective gap, normalized random blocks, and function-call accounting. For μ, we use the paper-style rule with a lower numerical floor 10^{-8}."

中文：解释 metric 和 μ：主要看 relative gap；random method 用 blocks of n directions；μ 有数值下限避免过小。

Mastery checklist / 掌握检查

- EN: Can write the quadratic chain from memory or recognize each term.
- EN: Can define GM, RG_μ, and RG_0.
- EN: Can explain why relative gap is used.
- 中文：能说清楚一个 random block $= n$ 个随机方向，而不是一次 function call。

Slide 12. Smooth Quadratic Experiment: Results / 光滑二次实验：结果

Owner: Member B.
Final PDF overlay pages: 49-53.
Paper source: Sec. 8.1, original PDF pp. 28-30, journal pp. 554-556, especially Table 1 for simple random search; Sec. 8.4, original PDF p. 36, journal p. 562, for interpretation.
Key formulas: function (70), method RG_μ (54), rate statement (56).
Important note: The numeric values in this slide are our reproduction-style experiment, not the original paper table.

Content task / 页面任务

EN: Interpret the plot and table carefully. The main result is not a formal dimension-scaling proof; it is qualitative support for the paper' s dimension-cost story.

中文：谨慎解释图表。核心不是证明 scaling，而是支持 "random gradient-free methods pay dimension cost" 的论文主线。

Overlay notes / 按 overlay 讲稿

Overlay 1 - plot only

EN: "For $n = 64$, the full-gradient method drops below the 10^{-3} target very early on the normalized budget axis. RG_μ decreases steadily but much more slowly."

中文：先只讲左图。GM 很快过阈值；RG_μ 稳定下降但慢很多。

Overlay 2 - table appears

EN: "The table summarizes normalized budget to reach relative gap 10^{-3} across dimensions. RG rows count blocks of n random directions. GM rows report iterations divided by n, so this is a normalized comparison, not direct function-call equality."

中文：解释表格口径。RG 是 n 个方向为一 block；GM 是 iterations divided by n。不能直接把二者看成完全相同成本。

Overlay 3 - first bullet: GM earlier

EN: "The first takeaway is simple: when the full gradient is available, GM reaches 10^{-3} much earlier."

中文：第一条结论：梯度可用时，GM 更快。

Overlay 4 - second bullet: $\mathbf{RG}_\mu$ tracks $\mathbf{RG}_0$

EN: "Second, RG_μ nearly matches diagnostic RG_0, so in this smooth quadratic test, finite-difference bias is small relative to the random-direction effect."

中文：第二条结论：RG_μ 与 RG_0 很接近，说明这里 finite-difference bias 很小。

Overlay 5 - third bullet: dimension-cost story, not proof

EN: "Third, because one RG block uses n random directions, the result supports the dimension-cost story. But it is not a proof: it is a small empirical check with finite dimensions, finite seeds, and a fixed test problem."

中文：第三条结论要谨慎：支持维度代价，但不是理论证明。

Specific table interpretation / 表格解释准备

EN: If asked why the numbers are not monotone in n, answer: "This table is not designed as a full scaling law estimate. Step size, target level, random seed behavior, and finite experiment horizon can all affect the normalized budget. The robust conclusion is the qualitative separation between GM and RG methods, and the close tracking of RG_μ and RG_0."

中文：如果被问"为什么不同维度下 RG budget 不单调"，回答：这个表不是严格 scaling law 估计，只是小样本 reproducibility check；步长、阈值、随机性、实验预算都会影响数值。稳健结论是 GM 与 RG 的差距，以及 RG_μ 与 RG_0 接近。

Mastery checklist / 掌握检查

- EN: Can explain each axis of the plot.
- EN: Can explain "normalized budget to 10^{-3}".
- EN: Can explain why RG_μ rows count blocks of n random directions.
- 中文：不要说"实验证明了 $O(n)$"。只能说"实验支持这个 dimension-cost story"。

Slide 13. Final Takeaways and Limitations / 总结与局限

Owner: Member B closes; Leader can answer theoretical Q&A.
Final PDF overlay pages: 54-58.
Paper source: Abstract, original PDF p. 1, journal p. 527; Sec. 1.2, original PDF p. 5, journal p. 531; Sec. 8.4, original PDF p. 36, journal p. 562.
Key formulas: summarized rates from Sections 4-7: nonsmooth $O(\sqrt{n/k})$-type, smooth $O(n/k)$, accelerated $O(n^2/k^2)$, nonconvex stationarity.

Content task / 页面任务

EN: Close the talk by summarizing the paper's conceptual contribution and practical limitation.

中文：用一页收束全文：理论贡献是什么，实际限制是什么。

Overlay notes / 按 overlay 讲稿

Overlay 1 - smoothing takeaway

EN: "The first takeaway is the theoretical bridge: Gaussian smoothing makes finite-difference random vectors estimate ∇f_μ in expectation."

中文：第一条：Gaussian smoothing 是函数值方法和梯度方法之间的桥。

Overlay 2 - dimension-cost takeaway

EN: "The second takeaway is that random gradient-free methods mimic first-order methods, but usually with an extra dimension cost."

中文：第二条：random gradient-free 模仿 first-order methods，但通常多维度代价。

Overlay 3 - acceleration takeaway

EN: "Acceleration is still possible. In the smooth convex case, the accelerated random method keeps the $1/k^2$ shape, though with n^2 dependence."

中文：第三条：可以加速，但加速只改善 k，不消除维度惩罚。

Overlay 4 - when to use

EN: "Therefore, these methods are best used when gradients are unavailable, unreliable, too memory-heavy, or too expensive to implement."

中文：第四条：什么时候值得用？当梯度不可得、不可靠、AD 成本高、代码实现太贵时。

Overlay 5 - summary table and limitations

EN: "The table summarizes the rate shapes. The limitation is also clear: if accurate gradients are available, gradient methods are usually faster; random gradient-free methods are sensitive to dimension and the smoothing radius μ."

中文：最后用表格总结 rate。局限：高维更慢，对 μ 敏感；如果梯度可得，通常不应优先用 random gradient-free。

Final closing line / 结束语

EN: "So the paper does not claim that zero-order methods dominate gradient methods. Its contribution is showing that when gradients are impractical, random Gaussian finite differences can recover first-order-like behavior with explicit complexity bounds."

中文：最终收束：论文不是说 zeroth-order 比 gradient methods 更好，而是当梯度不可用时，它给出了可证明的替代方案。

Mastery checklist / 掌握检查

- EN: Can summarize the entire talk in three sentences.
- EN: Can state the limitation without weakening the paper's contribution.
- 中文：能同时说清楚"有价值"和"不应滥用"。

---

3. Backup slides / 备用页指导

Backup slides should not be presented in the main 20-minute flow unless the professor asks a question. Each backup slide has one owner who should be ready to explain it in 30-60 seconds.

备用页不进入正式 20 分钟，除非教授提问。每张备用页应准备 30-60 秒解释。

Backup B1. Unified Notation / 统一记号

Owner: Shared; Leader maintains consistency.
Final PDF overlay page: 59.
Paper source: Sec. 1.3 notation, original PDF pp. 5-6, journal pp. 531-532; Sec. 2-3 definitions.
Key formulas: norm definitions in Sec. 1.3, smoothing (9), oracles (30).

Use case / 使用场景

EN: Use this if someone asks what L_0, L_1, R, g_μ, or π_Q means.

中文：如果有人问某个符号是什么意思，就翻到这页。

Backup notes / 讲法

EN: "For the talk we use the Euclidean case $B = I$. The paper is more general and writes the oracle with Bu and the update with $B^{-1}g$. The main concepts are unchanged."

中文：说明 slides 采用 $B = I$ 简化；论文中用一般几何结构。

Checklist / 检查

- All speakers must use the same notation: RG_μ, RG_0, RS_μ, RS_0.
- 不要混用 O 和 0；RG_0 是数字零。

Backup B2. Gaussian Gradient Identity / Gaussian 梯度恒等式

Owner: Leader.
Final PDF overlay page: 60.
Paper source: Sec. 2, original PDF pp. 8-10, journal pp. 534-536.
Key formulas: integral form before (21), gradient identity (21), symmetric form (26).

Use case / 使用场景

EN: Use this if asked "why is the expectation of the finite-difference vector equal to ∇f_μ?"

中文：如果被问"为什么有限差分向量的期望就是 ∇f_μ"，用这页。

Backup notes / 讲法

EN: "Write f_μ as a Gaussian integral centered at x. Change variables from u to $y = x + \mu u$, differentiate under the integral, and then use $E[u] = 0$ to subtract $f(x)$ inside the expectation. That gives the finite-difference form."

中文：讲三个步骤：换元、对 x 微分、利用 $E[u] = 0$ 减去 $f(x)$。不需要现场推所有积分常数。

Checklist / 检查

- Know that this identity is for f_μ, not necessarily f.
- 能解释为什么可以把 $f(x)$ 减进去：因为 $E[u] = 0$。

Backup B3. Accelerated Method Skeleton / 加速方法骨架

Owner: Leader.
Final PDF overlay page: 61.
Paper source: Sec. 6, original PDF pp. 22-25, journal pp. 548-551.
Key formulas: Method FG_μ (60), Theorem 9 (62).

Use case / 使用场景

EN: Use this if someone asks for more details about the accelerated algorithm.

中文：如果有人问 accelerated method 的具体结构，用这页。

Backup notes / 讲法

EN: "The method is structurally a Nesterov-type method: maintain x_k and v_k, form y_k, query $g_\mu(y_k)$, and update both sequences. The main slide intentionally keeps the parameter algebra out of the flow."

中文：强调算法结构，而不是参数代数。参数式在论文 Method (60)，主汇报不展开。

Checklist / 检查

- Can point to the four-step skeleton.
- Can state the rate again: $O(n^2 L_1 R^2/k^2)$ plus smoothing error.

Backup B4. Function Call Accounting / 函数调用计数

Owner: Member B.
Final PDF overlay page: 62.
Paper source: Sec. 3 oracle definitions, original PDF p. 10, journal p. 536; Sec. 8.1 tables, original PDF pp. 29-30, journal pp. 555-556.
Key formulas: oracle definitions (30); RG_μ method (54). Paper Table 1 reports random methods in blocks of n iterations.

Use case / 使用场景

EN: Use this if someone asks whether random blocks, function calls, and gradient evaluations are directly comparable.

中文：如果有人问"GM 和 RG 的成本能不能直接比较"，用这页。

Backup notes / 讲法

EN: "A forward finite-difference oracle usually needs $f(x)$ and $f(x + \mu u)$, though $f(x)$ may be cached. The symmetric oracle needs two shifted values. The paper often reports random methods in blocks of n random iterations. A gradient evaluation is not universally equal to one function call."

中文：说明成本口径：forward finite difference 通常两个函数值，symmetric 两个 shifted values；random methods 常用 n 次迭代为一个 block；gradient evaluation 成本因问题而异。

Checklist / 检查

- Do not claim one gradient evaluation equals one function call.
- Do say that the paper' s qualitative comparison is usually dimension-cost based.

Backup B5. Dimension-Scaling Summary / 维度 scaling 总结

Owner: Member B.
Final PDF overlay page: 63.
Paper source: Sec. 8.1 original experiment tables, original PDF pp. 29-30, journal pp. 555-556; Sec. 8.4 conclusion, original PDF p. 36, journal p. 562.
Key formulas: not formula-based; based on experiment table and random-block accounting.

Use case / 使用场景

EN: Use this if someone asks for the full dimension summary behind Slide 12.

中文：如果有人问 Slide 12 的维度表详情，用这页。

Backup notes / 讲法

EN: "The table reports normalized budget to reach relative gap 10^{-3}. RG_μ rows count blocks of n random directions, while GM rows report iterations divided by n. We should not overinterpret monotonic scaling from this small table."

中文：说明表格不是完整 scaling law；只是支持维度成本主线。

Checklist / 检查

- Mention seeds 1:10 if reproducibility is asked.
- Mention that RG_0 is diagnostic and appears in detailed code/CSV, not as a main method recommendation.

---

4. Per-person final preparation checklist / 每位组员最终准备清单

Leader checklist / 组长清单

☐ Can explain the whole paper’ s logic in 60 seconds.
☐ Can derive or sketch Gaussian gradient identity from Backup B2.
☐ Can distinguish f, f_μ, ∇f, and ∇f_μ.
☐ Can explain why g_μ is a random oracle and why it is useful only in expectation.
☐ Can explain the $(n+4)$ second-moment bound conceptually.
☐ Can present acceleration without getting lost in Method (60) parameter algebra.
☐ Can answer why acceleration keeps $1/k^2$ but pays n^2.

中文检查：组长必须能讲清楚 smoothing -> oracle -> dimension factor -> acceleration 这条主线；如果教授追问证明，优先使用 Backup B2/B3。

Member A checklist / 组员 A 清单

☐ Can define the constrained nonsmooth problem $\min_{x\in Q} f(x)$.
☐ Can explain projection π_Q geometrically.
☐ Can compare RS_0, RS_μ, and subgradient method.
☐ Can explain smoothing bias $\mu L_0\sqrt{n}$.
☐ Can explain why finite-difference nonsmooth complexity is worse.
☐ Can explain stochastic objective $f(x) = E_\xi[F(x,\xi)]$.
☐ Can explain same-ξ finite difference.

中文检查：组员 A 不需要推完整 Theorem 5-7，但必须能说清楚每个算法的 problem, oracle, update, rate shape。

Member B checklist / 组员 B 清单

☐ Can state smooth RG_μ update and step size scaling.
☐ Can explain $O(1/k)$ versus $O(n/k)$.
☐ Can explain nonconvex stationarity instead of global optimality.
☐ Can write the quadratic-chain objective and explain why gaps are computable.
☐ Can define GM, RG_μ, and diagnostic RG_0.
☐ Can explain the plot axes and table in Slide 12.
☐ Can explain random-block accounting and why the experiment is not a proof.
☐ Can close with both value and limitation of zero-order methods.

中文检查：组员 B 要对实验细节最熟，尤其是 normalized budget、blocks of n random directions、RG_μ/RG_0 的区别。

---

5. Suggested 20-minute timing / 建议时间分配

Slide	Owner	Time	Notes
1	Leader	0:40	Title and role table
2	Leader	1:20	Motivation
3	Leader	1:50	Gaussian smoothing
4	Leader	1:45	Oracles and dimension factor
5	Leader	1:45	Acceleration

Slide	Owner	Time	Notes
6	Member A	1:20	Nonsmooth projected search
7	Member A	1:20	Bias and finite differences
8	Member A	1:10	Stochastic extension
9	Member B	1:30	Smooth baseline
10	Member B	1:10	Nonconvex extension
11	Member B	1:30	Experiment setup
12	Member B	2:00	Experiment results
13	Member B	1:30	Final takeaways
Total		approx. 20:00	Backup excluded

---

6. Final consistency rules / 最终一致性规则

1. Always write $\mathbf{RG}_0$ with a zero, not the letter O.
   一律写 RG_0，不要写 RG-O。
2. Say **finite-difference oracle estimates ∇f_μ in expectation**, not simply "estimates ∇f" in all cases.
   要说估计 ∇f_μ，不要泛泛说估计 ∇f。
3. Say **dimension cost** or **dimension dependence**, not "dimension curse" unless explaining informally.
   建议使用 dimension cost / dimension dependence，不要过度夸张成 curse。
4. Say **the experiment supports the story; it is not a proof**.
   实验是支持，不是证明。
5. Do not claim gradient-free methods are better when gradients are cheap and accurate.
   不要说 gradient-free 比 gradient method 更好；应说在梯度不可得或成本太高时有价值。
6. The main narrative should remain:
   Motivation -> Gaussian smoothing -> random oracle -> dimension cost -> problem classes -> experiment -> limitations.
   主线应保持：动机 -> Gaussian smoothing -> 随机 oracle -> 维度代价 -> 各类问题 -> 实验 -> 局限。

---

7. One-sentence role summaries / 每人一句话总结

Leader EN: "I explain the theoretical mechanism: Gaussian smoothing turns function-value differences into random gradient estimates, and the oracle variance creates the dimension cost, even under acceleration."

组长中文："我负责理论机制：Gaussian smoothing 如何把函数值差分变成随机梯度估计，以及 oracle 方差如何带来维度代价，即使加速也不能完全消除。"

Member A EN: "I explain how the same random-search machinery applies to nonsmooth constrained problems and stochastic objectives, where projection, smoothing bias, and sample noise matter."

组员 A 中文："我负责非光滑约束和随机目标函数，重点是投影、smoothing bias 和样本噪声。"

Member B EN: "I explain the smooth baseline, the nonconvex stationarity shift, and the Julia experiment showing the practical dimension-cost story."

组员 B 中文："我负责光滑基线、非凸 stationarity 变化，以及 Julia 实验如何支持维度成本主线。"